

How the KT-D1V5 Shock-Tube Solver Works

A plain-language guide to the math and physics

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The short version. The solver does not compute density, velocity and pressure directly. It tracks five groups of imaginary gas particles, each moving at a fixed speed. In every time step the particles

1. **move** left or right through small cells of the tube, and
2. **collide**, which nudges each group toward a “balanced” amount (the equilibrium).

Adding the groups up in the right ways gives the density, velocity, temperature and pressure of the gas. This guide explains each piece, then shows how well the result matches the exact answer for the Sod shock tube.

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1 The problem: the Sod shock tube

Picture a long tube, 20 units long, with a thin wall in the middle ($x = 10$). The left side holds dense, high-pressure gas and the right side holds thin, low-pressure gas:

$$\text{left: } \rho = 1, u = 0, p = 1 \qquad \text{right: } \rho = 0.125, u = 0, p = 0.1.$$

Here ρ is density, u is flow velocity and p is pressure. At time zero the wall disappears. By the final time $t = 0.15$ three waves have formed (fig. 1):

- a **rarefaction** (expansion fan) moving left, where the gas smoothly thins out;
- a **contact**, the boundary between the “old left gas” and the “old right gas”. Density and temperature jump here, but pressure and velocity do not;
- a **shock** moving right, where all four quantities jump.

This problem is a standard test because its exact answer is known [8, 10]. The exact values used for comparison are listed in table 1. The main ones are a middle-region pressure $p^* = 0.3031$, a flow speed $u^* = 0.9275$, and a shock speed of 1.752.

Table 1: Exact Sod values at $t = 0.15$ (gas with $\gamma = 1.4$).

Quantity	Value	Meaning
p^*, u^*	0.3031, 0.9275	middle-region pressure and speed
ρ left / right of contact	0.4263 / 0.2656	density jumps at the contact
T left / right of contact	0.711 / 1.141	temperature jumps at the contact
Shock speed	1.752 (Mach 1.66)	shock is 0.263 right of the wall
Contact position	0.139 right of the wall	moves with the gas
Rarefaction	0.177 to 0.011 left of wall	smooth expansion

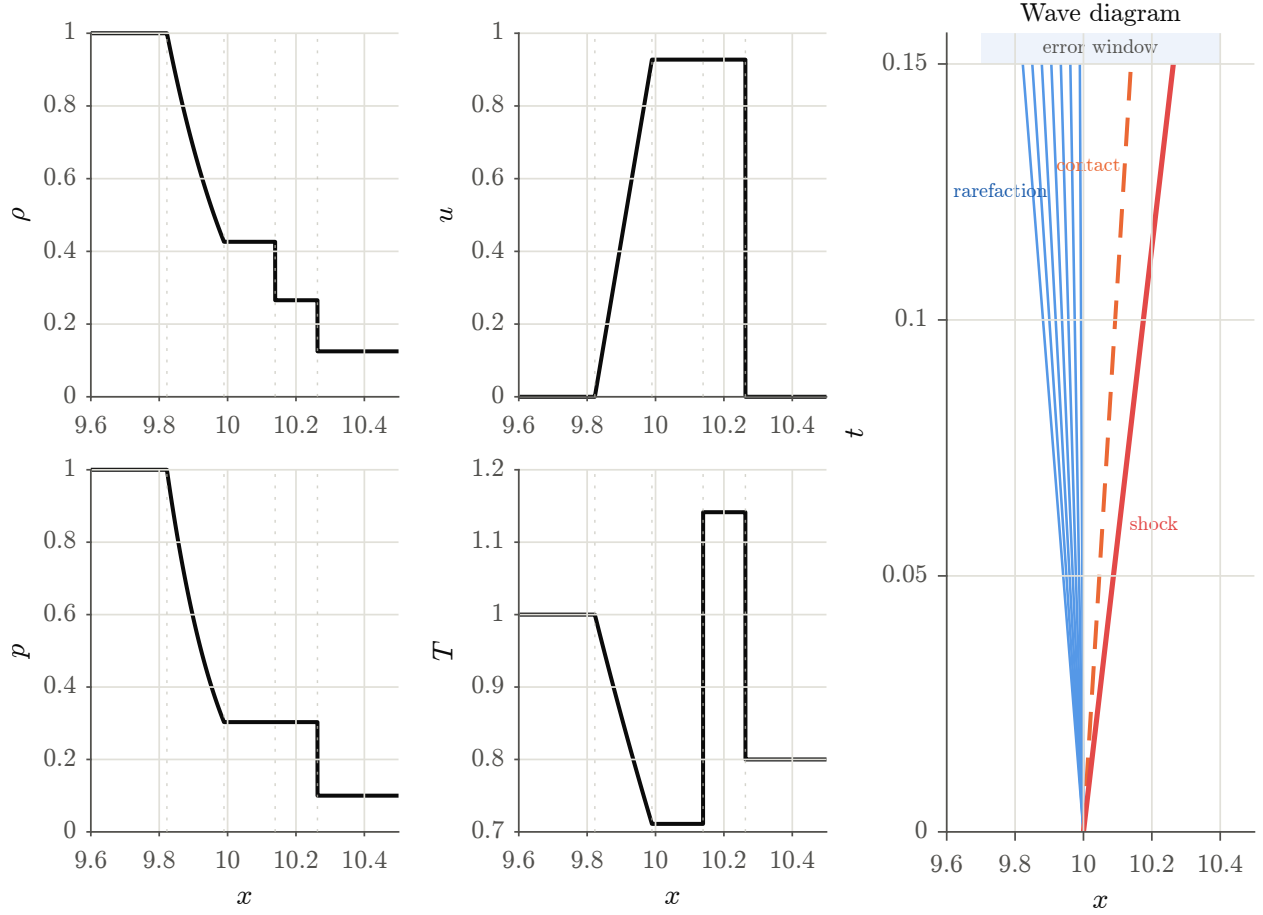


Figure 1: The exact answer at $t = 0.15$. Left: density, velocity, pressure and temperature along the tube (dotted lines mark the waves). Right: the paths of the waves in time. The shaded strip is the region used to measure errors.

2 The physics idea: particles instead of fluid

2.1 The Boltzmann–BGK equation

Gas is made of molecules moving in all directions. Kinetic theory describes the gas by a *distribution* f : how many molecules are at each place moving at each speed. The simplest useful model of how f changes is the BGK equation [1]:

$$\underbrace{\frac{\partial f}{\partial t}}_{\text{change in time}} + \underbrace{c \frac{\partial f}{\partial x}}_{\text{particles moving}} = \underbrace{\frac{f^{\text{eq}} - f}{\tau}}_{\text{collisions}}. \quad (1)$$

The left side says that particles with speed c simply drift along the tube. The right side says that collisions pull f toward a balanced shape f^{eq} (the *equilibrium*). The **relaxation time** τ sets how fast that happens.

What τ means. τ is roughly the time between collisions. A small τ means the gas re-balances almost instantly and behaves like an ideal (frictionless) fluid. A larger τ adds a little “stickiness” (viscosity) that smooths sharp jumps. This solver uses $\tau = 5 \times 10^{-6}$, which is very small.

When τ is small, averaging (1) over all particles gives back the ordinary **Euler equations** of gas dynamics (conservation of mass, momentum and energy) [2]. That is why a particle model can solve a fluid problem.

2.2 Only five particle speeds: the D1V5 model

A real gas has every speed. The solver keeps only **five** (fig. 2), following Kataoka and Tsutahara [4]:

$$c \in \{-3, -1, 0, +1, +3\}.$$

“D1V5” means one dimension, five velocities. The group with speed 0 does not move, but it carries an extra energy “backpack” ($\eta_0 = 1.75$). That backpack stands in for the energy a real molecule stores in spinning and in sideways motion, which is what lets the model represent air ($\gamma = 1.4$).

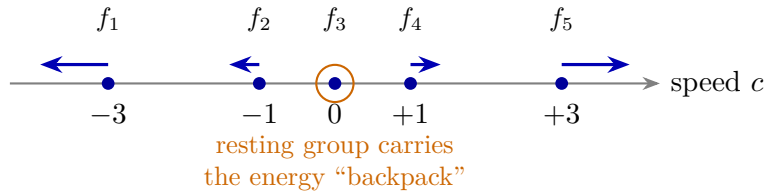


Figure 2: The five particle groups. Arrows show their direction and relative speed.

2.3 Getting density, velocity, temperature and pressure back

Each cell stores five numbers, $f_1, \dots, f_5$. The gas properties come from weighted sums of them:

$$\begin{aligned}\rho &= f_1 + f_2 + f_3 + f_4 + f_5 && \text{(total amount = density)} \\ \rho u &= \sum_i c_i f_i && \text{(speed-weighted sum = momentum)} \\ \rho(5T + u^2) &= \sum_i (c_i^2 + \eta_i^2) f_i && \text{(energy)} \\ p &= \rho T && \text{(ideal-gas law)}\end{aligned}$$

Temperature is found last: take the total energy and subtract the part that is just bulk motion (u^2). Because it is a difference of two large numbers, T is the most sensitive result. A small error in one group (especially a fast group) shows up in T more strongly than in ρ or p . This is why temperature shows overshoots first (section 6.5).

2.4 The equilibrium

The balanced state f^{eq} is not guessed. Its five values are chosen so that they give back exactly the right density, momentum and energy, and also the right *flow* of momentum and energy that the Euler equations need. Those are five conditions for five unknowns, so there is exactly one answer. It is the formula in the solver's equilibrium function. I checked it numerically on 2000 random gas states; it reproduces all five quantities to about 10^{-15} .

Two facts worth knowing:

- Some equilibrium values are **negative** (the slow groups, for the temperatures in this problem). That is allowed: the groups are bookkeeping quantities, not real particle counts. It is also why the solver checks that ρ and T stay positive rather than checking each f_i .
- Mass, momentum and energy are **never created or destroyed** by collisions, because the equilibrium is built from the same totals as f .

3 How the computer solves it

The solver repeats the same step many times until $t = 0.15$. Each step has three ingredients.

3.1 Ingredient 1: cells and fluxes (finite volumes)

The tube is cut into N_x equal cells of width $\Delta x = 20/N_x$. For example, $N_x = 50,000$ gives $\Delta x = 0.0004$. Each cell stores the average f_i inside it. A cell's value changes only because of what flows in and out through its two walls, plus collisions:

$$\text{change in cell} = -\frac{\text{flux out right wall} - \text{flux in left wall}}{\Delta x} + \frac{f^{\text{eq}} - f}{\tau}.$$

Whatever leaves one cell enters its neighbour, so nothing is lost. The results confirm this: total mass and energy stay constant to about 10^{-12} .

3.2 Ingredient 2: upwinding, and MUSCL for sharper answers

To find the flux through a wall, you need the value of f_i right at that wall. Because each group moves in one direction only, the rule is simple (fig. 3):

- right-moving groups ($c = +1, +3$) take the value from the cell on the *left* of the wall;
- left-moving groups ($c = -1, -3$) take it from the cell on the *right*;
- the resting group ($c = 0$) never crosses a wall.

This is called **upwinding**: you look “upstream” for the information.

Using the plain cell average at the wall (“first-order”) is stable but smears every jump over many cells. **MUSCL** [11] does better. It draws a sloped line through each cell and reads the wall value off that line:

$$\text{value at right wall of cell } j = f_j + \frac{1}{2} \sigma_j,$$

where σ_j is the slope (the change across one cell).

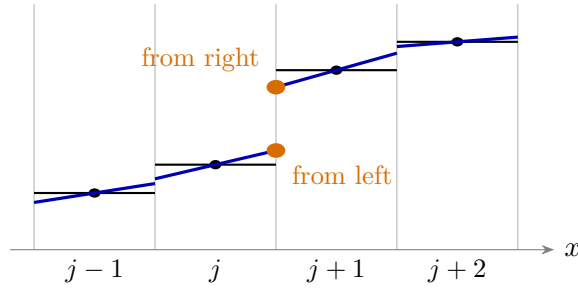


Figure 3: MUSCL. Black: cell averages. Blue: sloped lines inside each cell. Orange: the two possible values at the wall between cells j and $j+1$. Right-moving groups use the “from left” value; left-moving groups use the “from right” value.

3.3 Ingredient 3: the slope limiter (generalized minmod)

A slope that is too steep near a jump creates fake wiggles (overshoots). A **limiter** decides how steep the slope may be. Let d_L be the change from the left neighbour and d_R the change to the right neighbour. The solver uses the **generalized minmod** limiter [5]:

$$\sigma = \minmod\left(\theta d_L, \frac{1}{2}(d_L + d_R), \theta d_R\right), \quad \theta = 1.2.$$

“minmod” means: if all three candidates have the same sign, pick the one closest to zero; if the signs disagree (a peak or a dip), use a slope of zero. So:

- at a peak or dip the slope is switched off, which prevents new wiggles;
- in smooth regions the slope is the smallest of three reasonable estimates;
- θ is a dial. $\theta = 1$ is plain *minmod* (very safe, but smears jumps). $\theta = 2$ is *MC* (sharp, but overshoots). $\theta = 1.2$ sits in between.

Theory [3, 9] shows this family never creates new wiggles in a single group’s motion as long as $1 \leq \theta \leq 2$, which is why θ is kept in that range (fig. 4).

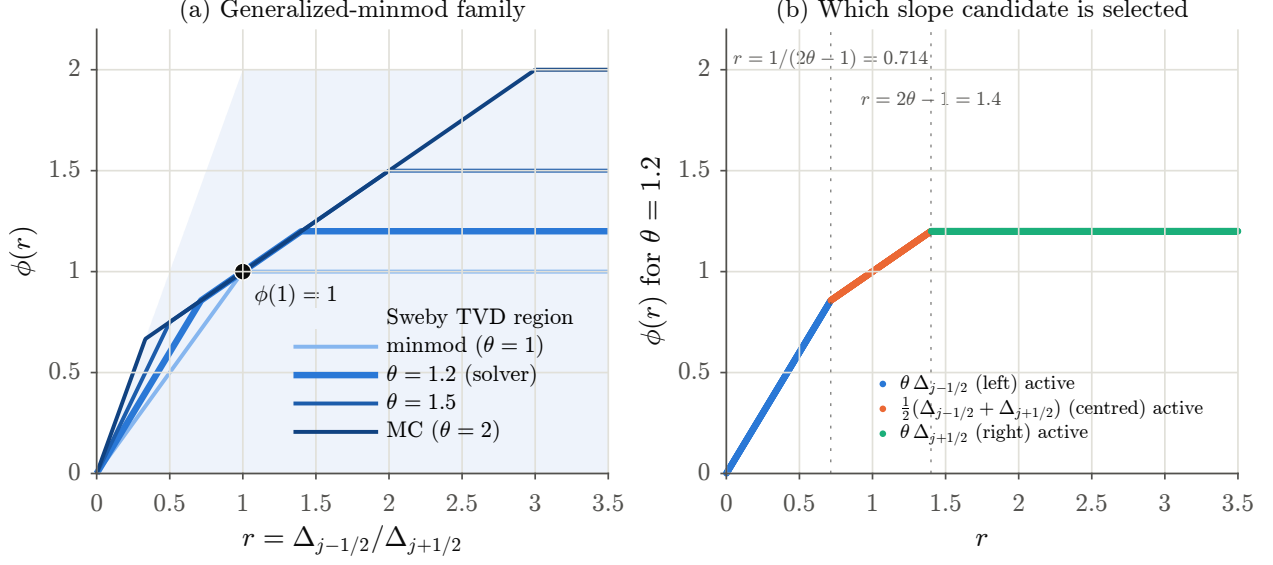


Figure 4: (a) How steep the slope is allowed to be for different θ . The horizontal axis measures how smooth the data are ($r = 1$ means perfectly smooth). All curves stay inside the shaded “no-new-wiggles” region. (b) Which of the three candidates $\theta = 1.2$ picks.

3.4 Moving forward in time: SSP-RK3

Instead of one big jump per step, the solver takes three smaller “looks” and blends them (the SSP Runge–Kutta method of Shu and Osher [7]). Writing $L(f)$ for “movement plus collisions”:

$$\begin{aligned} f^{(1)} &= f^n + \Delta t L(f^n) \\ f^{(2)} &= \frac{3}{4} f^n + \frac{1}{4} [f^{(1)} + \Delta t L(f^{(1)})] \\ f^{n+1} &= \frac{1}{3} f^n + \frac{2}{3} [f^{(2)} + \Delta t L(f^{(2)})] \end{aligned}$$

Each line is an average of simple forward steps. That keeps the good “no new wiggles” behaviour of the limiter while being much more accurate than a single forward step.

How big is each time step? Two limits are checked and the smaller one is used:

$$\Delta t = \min \left(\underbrace{\text{CFL} \cdot \frac{\Delta x}{3}}_{\text{movement limit}}, \underbrace{\tau/4}_{\text{collision limit}} \right).$$

The movement limit keeps the fastest group from jumping too far in one step. The collision limit keeps the relaxation accurate. With $\tau = 5 \times 10^{-6}$, the collision limit gives $\Delta t = 1.25 \times 10^{-6}$, and that is smaller than the movement limit on *every* grid used here, even with $\text{CFL} = 0.025$. So every run used the same time step and the same 120,000 steps. In practice the CFL setting never controlled the step.

Safety checks. After each stage the solver checks that density and temperature are still positive. If not, it halves the time step and, if needed, switches to a safer limiter. In every run reported here this *never happened*: zero retries, zero fallbacks.

4 What τ does to the answer

The exact Sod answer assumes a perfectly frictionless gas ($\tau = 0$). The solver uses a small but nonzero τ , so even a perfect computer would give slightly smoothed jumps. Working through the math for this five-speed model shows the following.

- The **shock** is smoothed only over about $7\tau \approx 0.00004$, which is much thinner than one cell. The shock’s width in the results comes from the grid, not from τ .
- The **contact** keeps spreading with time, like heat diffusing. By $t = 0.15$ its natural width is about 0.0012–0.0015, roughly three cells on the finest grid.
- The spreading rate at the contact depends on how fast the gas is moving there. At the Sod contact ($u \approx 0.93$) it is about 8 times slower than for gas at rest. If the gas moved faster than the slow particle speed ($u > 1$), the model would break down. So this velocity set is only suitable for flows slower than that.

The practical meaning: refining the grid can only help until the cells are smaller than this natural contact width. For $\tau = 5 \times 10^{-6}$ that happens around $N_x \approx 200,000$ cells. For the older $\tau = 5 \times 10^{-5}$ it happens around 50,000 cells, which explains why the earlier study stopped improving there.

5 How accuracy is measured

At each cell the numerical value is compared with the exact value.

- **L_1 error:** the average size of the miss over all cells.
- **L_2 error:** the root-mean-square miss. It punishes large local misses more than L_1 does.
- **Temperature peak excess:** highest computed temperature minus the highest exact temperature. A positive value means an overshoot.
- **Order of convergence p :** how fast the error shrinks when the cells get smaller. If halving Δx cuts the error in half, $p = 1$; if it cuts it by a quarter, $p = 2$. It is computed as

$$p = \frac{\log(E_{\text{coarse}}/E_{\text{fine}})}{\log(\Delta x_{\text{coarse}}/\Delta x_{\text{fine}})}.$$

What order to expect. The method is “second order” only where the solution is smooth. Sod has jumps, and every method smears a jump over a few cells, so the error there cannot shrink as fast [6]. For this kind of method the expected orders are about:

- L_1 : between 0.67 (from the contact) and 1 (from the shock);
- L_2 : between 0.33 (contact) and 0.5 (shock).

So an L_2 order around 0.4–0.5 is *normal and correct* here. It does not mean something is wrong.

6 Results

Settings for every run: $\tau = 5 \times 10^{-6}$, CFL = 0.025, $\theta = 1.2$, final time 0.15, tube length 20.

6.1 The solution looks right

Figure 5 shows the finest grid (50,000 cells) against the exact answer. All four quantities match closely with no wiggles. The errors are concentrated in a few cells at the contact and the shock, as expected. Figure 6 zooms in on each wave: every grid refinement makes the jumps sharper.

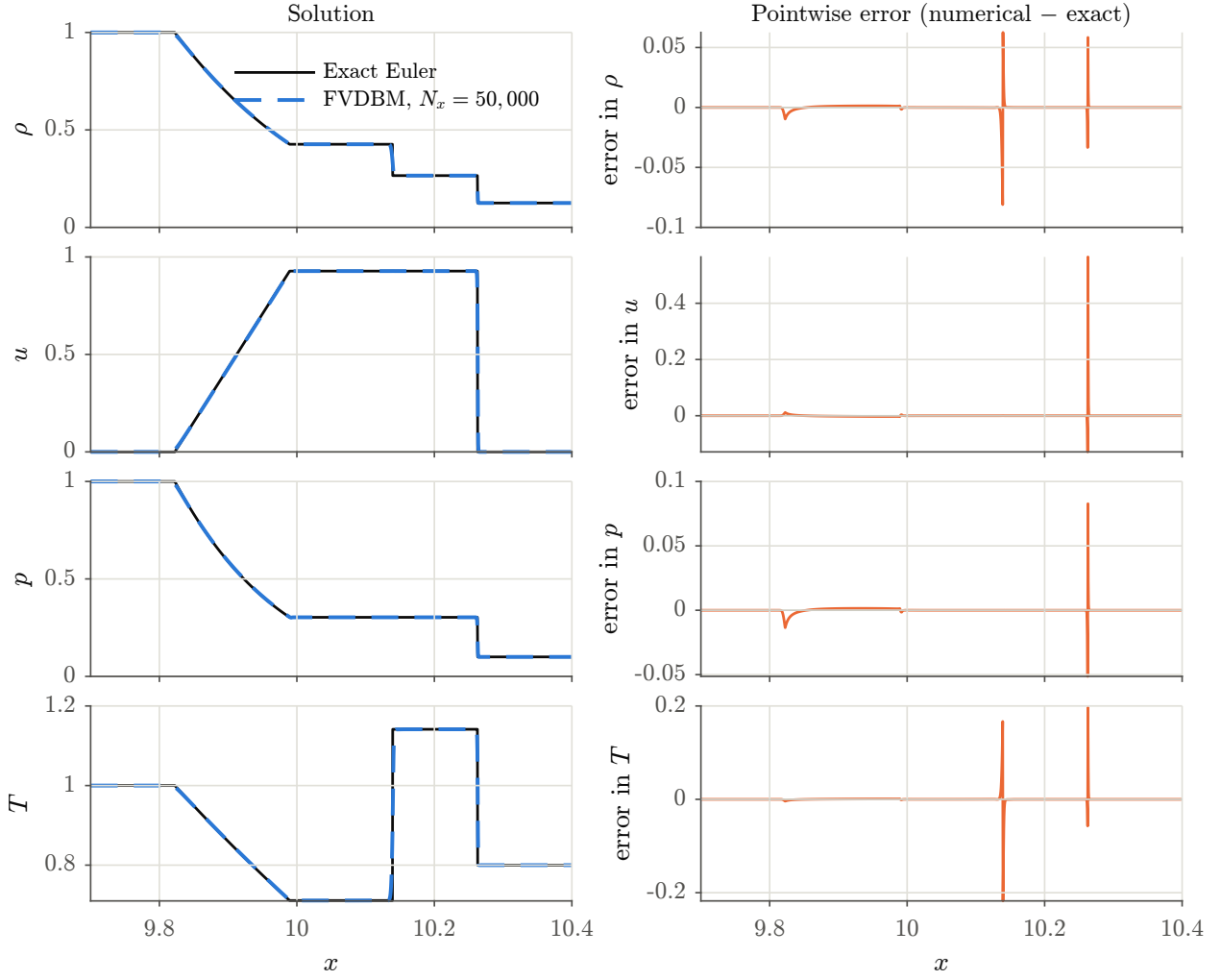


Figure 5: Finest grid (dashed blue) versus the exact answer (black), with the point-by-point error on the right.

6.2 Grid convergence

Table 2 and fig. 7 give the main result.

- Every error gets smaller as the grid is refined.
- The orders ($L_2 \approx 0.4\text{--}0.6$, $L_1 \approx 0.8\text{--}0.9$) are exactly in the expected range for a problem with a shock and a contact.
- All runs were clean: no retries, no fallbacks, and mass and energy were conserved to about 10^{-12} .

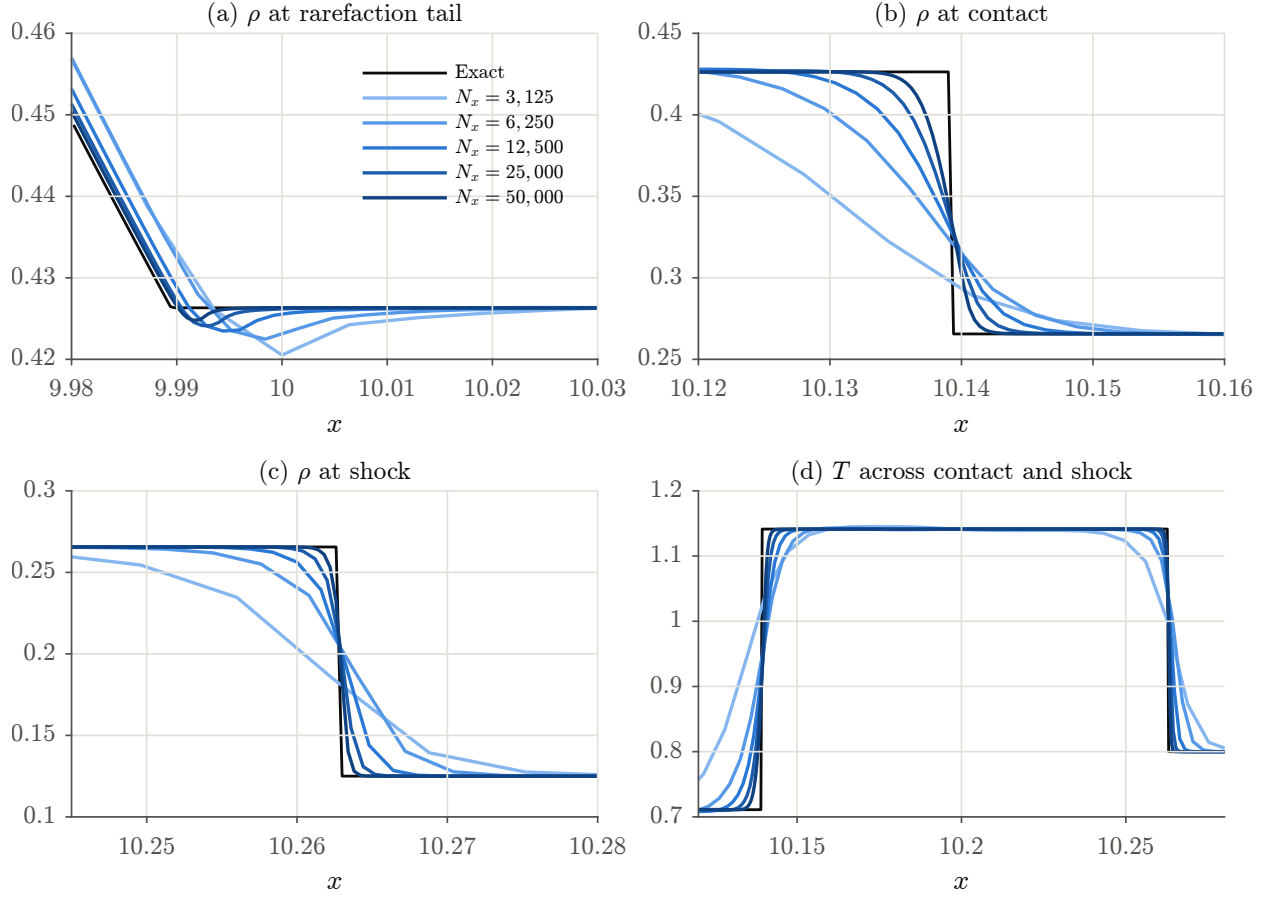


Figure 6: Close-ups at each wave. Darker blue means more cells. The shock stays about 3 cells wide. The contact is wider, and the small temperature overshoot on the coarsest grids disappears by 12,500 cells.

Table 2: Whole-tube L_2 errors and temperature peak excess.

Cells N_x	ρ	u	p	T	T peak excess
3,125	0.00316	0.00779	0.00308	0.00592	+0.0039
6,250	0.00181	0.00806	0.00164	0.00465	+0.0016
12,500	0.00139	0.00613	0.00114	0.00354	≈ 0
25,000	0.00100	0.00271	0.00072	0.00244	≈ 0
50,000	0.00078	0.00275	0.00051	0.00202	≈ 0
Order (L_2)	0.49	0.46	0.63	0.40	
Order (L_1)	0.83	0.89	0.88	0.81	

- The **contact** is the slowest part to improve. It limits the density and temperature errors. The shock stays about 3 cells wide at every resolution, which is typical.

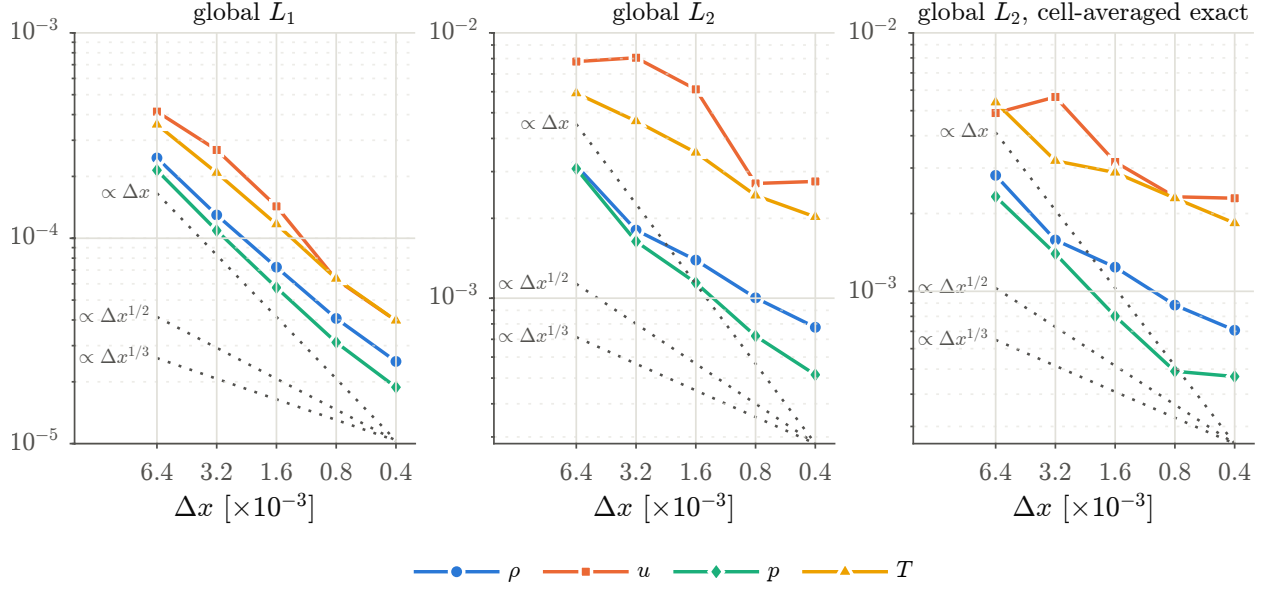


Figure 7: Errors versus cell size (smaller cells to the right). Dotted lines show reference slopes. Lines running parallel to the $\Delta x^{1/3}$ to $\Delta x^{1/2}$ guides are the expected behaviour.

6.3 Why the velocity error stopped improving from 25k to 50k cells

Nearly all of the velocity error sits in the few cells right at the shock, where the velocity drops from 0.93 to 0. How big that error is depends on *where the exact shock happens to fall inside a cell*: near a cell's edge or near its middle. That position is different on every grid. On the 25k grid the shock sits near the middle of a cell (smaller error); on the 50k grid it sits near an edge (larger error). Comparing grids where the shock sits in the same spot, the velocity error shrinks at order 0.50–0.51, exactly as expected (fig. 8). So the stall is a quirk of how the error is measured, not a problem with τ or the time step. The L_1 velocity error, which is less sensitive to this, improves steadily.

6.4 Effect of the smaller τ

Table 3 compares this study with the earlier one at a 10 times larger τ . With the old τ the error levelled off near 1.13×10^{-3} from 40,000 cells on. With the new τ it keeps dropping, reaching 0.78×10^{-3} at 50,000 cells. This matches the prediction above: the old τ made the contact naturally wider, so the grid could no longer help. (The earlier runs also used a larger time step, so this comparison changes two things at once.)

6.5 Comparing the limiters

All five limiters tried during the project were rerun at 6,250 cells with identical settings (table 4, fig. 9).

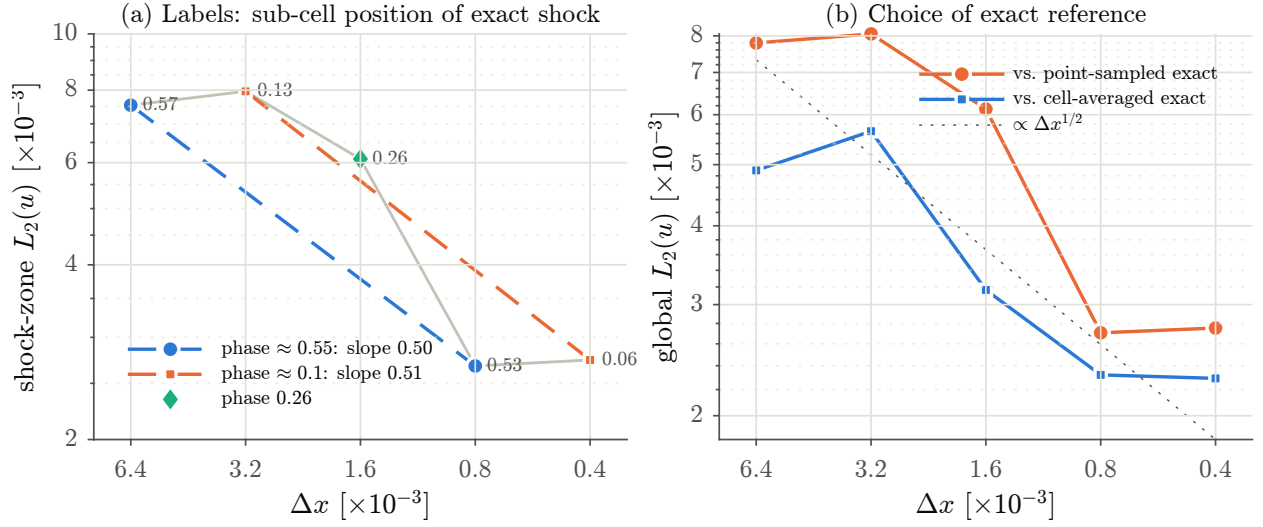


Figure 8: (a) Velocity error near the shock. The labels give where the exact shock falls inside its cell (0 = edge, 0.5 = middle). Grids with matching positions (same colour) improve at the expected rate. (b) The whole-tube velocity error.

Table 3: Density L_2 error at the old and new relaxation times.

Cells	old $\tau = 5 \times 10^{-5}$	new $\tau = 5 \times 10^{-6}$	new / old
3,125	0.00324	0.00316	0.97
6,250	0.00196	0.00181	0.92
12,500	0.00159	0.00139	0.87
25,000	0.00126	0.00100	0.80
40,000	0.00113	—	—
50,000	0.00113	0.00078	0.69

Table 4: Limiter comparison, 6,250 cells. Lower is better in both columns.

Limiter	average temperature error $L_1(T)$	temperature overshoot
Minmod	0.000265	0.00045
MC	0.000168	0.00672
Population hybrid	0.000216	0.00259
Macroscopic hybrid	0.000266	0.00062
Generalized minmod, $\theta = 1.2$	0.000208	0.00155

- **MC** has the smallest average error but by far the largest temperature overshoot.
- **Minmod** has almost no overshoot but the largest average error.
- **Generalized minmod** ($\theta = 1.2$) is in between. Compared with minmod, it lowers every average error by 20–23%. Compared with MC, it cuts the overshoot by 77%.

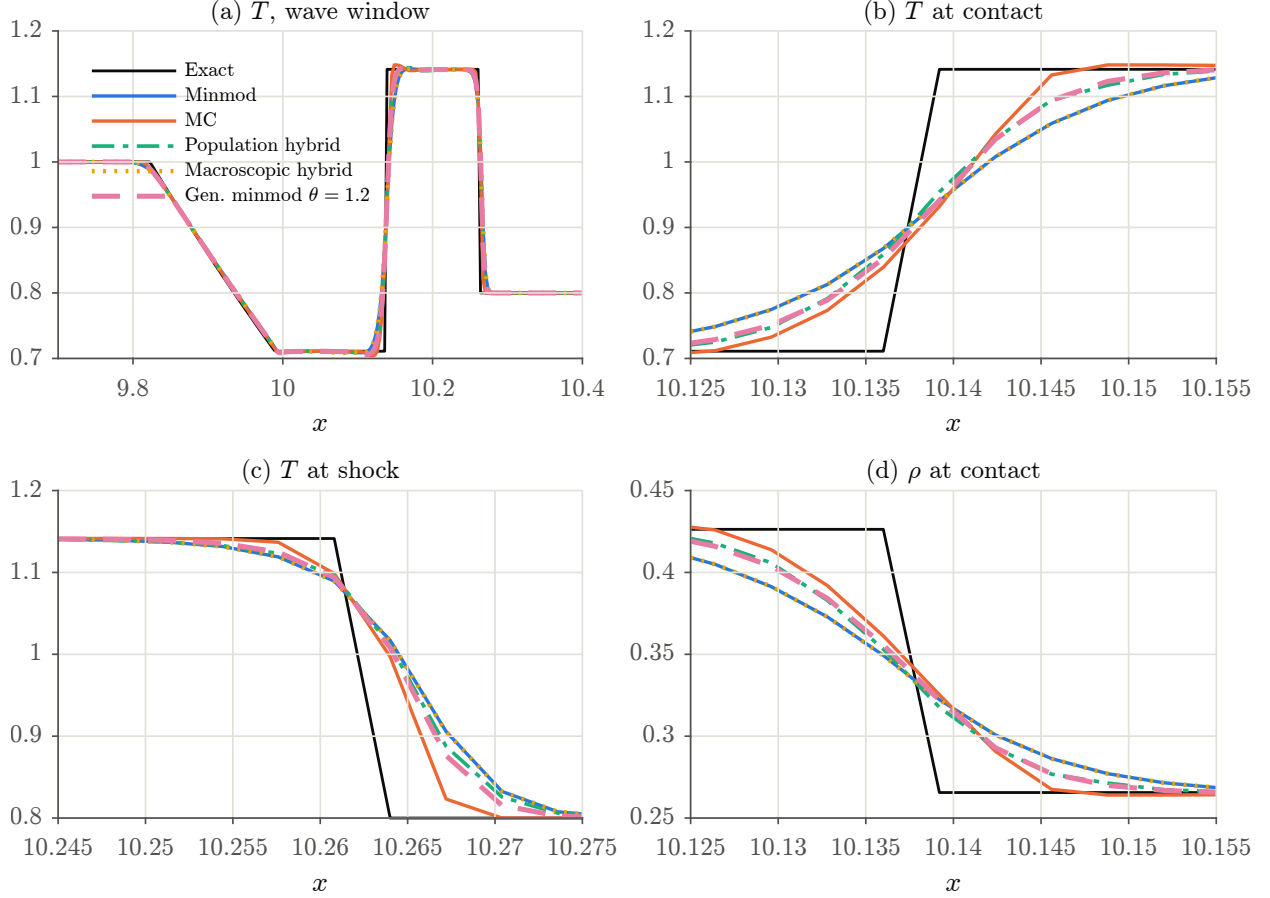


Figure 9: Temperature and density near the contact and shock for each limiter. MC (orange) is sharpest but overshoots just behind the contact.

6.6 Choosing θ

Running every θ from 1.0 to 2.0 (fig. 10) shows a clean tradeoff: as θ increases, the average error always goes down and the overshoot always goes up. No single θ wins on both. $\theta = 1.2$ gets about 59% of the possible error reduction while paying only about 18% of the possible overshoot. It is a sensible middle choice, not a “perfect” value. On finer grids the overshoot at $\theta = 1.2$ disappears anyway (table 2).

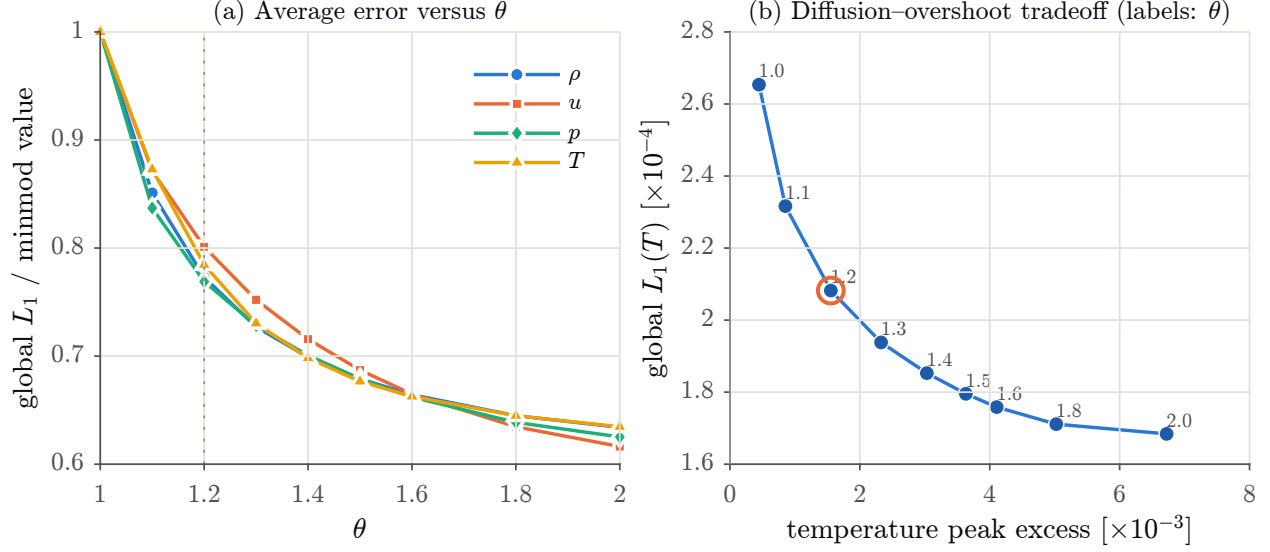


Figure 10: (a) Average error relative to minmod as θ changes. (b) Error versus overshoot; each dot is one θ , and the circle marks $\theta = 1.2$.

7 Key takeaways

1. The solver tracks five particle groups that move through cells and relax toward a balanced state. Their sums give density, velocity, temperature and pressure.
2. The five-speed model reproduces the Euler equations exactly when τ is small. τ adds a tiny amount of smoothing, mostly visible at the contact.
3. MUSCL with the generalized-minmod limiter makes jumps sharp without adding wiggles. SSP-RK3 steps forward in time accurately.
4. The errors shrink at the rates expected for a problem with a shock and a contact. All runs were stable and conserved mass and energy.
5. The velocity “stall” at fine grids is a measurement quirk (shock position inside a cell), not a physics problem.
6. $\theta = 1.2$ is a reasonable balance between smearing and overshoot.

Good next steps.

- Halve the time step once on the finest grid to confirm time error is small.
- Test on a smooth problem (for example a small sound pulse) to confirm true second-order accuracy, which Sod cannot show.
- Run a τ study at a fixed time step to test the contact-width prediction directly.
- Before moving to 2D, pick a velocity set whose slow speed is larger than the flow speeds you expect.

More detail. Full derivations, proofs and extra tables are in the companion technical report, *Mathematical and Physical Foundations of the KT-D1V5 Finite-Volume Discrete Boltzmann Solver* (d1v5_solver_mathematics.pdf).

Glossary

BGK collision A simple rule that pulls the particle groups toward equilibrium at a rate set by τ .

CFL number How far the fastest particles move in one time step, measured in cells.

Contact The boundary between two gases at the same pressure and speed but different density and temperature.

Equilibrium (f^{eq}) The balanced amounts of each particle group for a given density, velocity and temperature.

Finite volume Dividing the tube into cells and tracking what flows through each cell's walls.

Limiter A rule that caps the slope inside each cell to prevent fake wiggles.

MUSCL Using a sloped line in each cell, instead of a flat value, to get sharper results.

Rarefaction A smooth expansion where the gas thins out.

Relaxation time (τ) Roughly the time between collisions; smaller means closer to a frictionless gas.

Shock A thin, sudden jump in pressure, density, velocity and temperature.

SSP-RK3 A three-stage time-stepping method that is accurate and does not create wiggles.

Upwinding Taking information from the direction the particles are coming from.

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